

Differentiable functions – 9

Integral inequalities through derivatives

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Theoretical minimum.

Theorem 1 (Wirtinger). If $f \in C^1[0, \pi]$ with $f(0) = f(\pi) = 0$, then $\int_0^\pi f^2 \leq \int_0^\pi (f')^2$, with equality only for $f = c \sin x$. The periodic version has the same constant once the mean value is subtracted.

Theorem 2 (Poincaré and Hardy). For $f \in C^1[a, b]$ with $f(a) = 0$ one has $\int_a^b f^2 \leq \frac{(b-a)^2}{2} \int_a^b (f')^2$; and for f on $(0, \infty)$ with $f(0) = 0$, $\int_0^\infty \left(\frac{f(x)}{x}\right)^2 dx \leq 4 \int_0^\infty (f')^2$, where the constant 4 is sharp (Hardy).

Fact 1. Two techniques cover the whole sheet: integrate by parts to move a derivative from one factor to the other, and expand in the eigenfunctions of the corresponding boundary value problem — for $[0, \pi]$ with zero boundary conditions these are exactly $\sin nx$.

1. Prove Wirtinger's inequality in both ways: through the Fourier expansion in $\sin nx$, and by the substitution $f = u \sin x$ followed by an integration by parts. Determine the equality cases.
2. Prove the Poincaré inequality with the constant $\frac{(b-a)^2}{2}$ and then improve it to the sharp constant $\frac{(b-a)^2}{\pi^2}$ for functions vanishing at both endpoints.
3. Prove Hardy's inequality $\int_0^\infty \left(\frac{f}{x}\right)^2 \leq 4 \int_0^\infty (f')^2$ and show that the constant cannot be improved, exhibiting a minimising sequence.
4. Prove Opial's inequality: for $f \in C^1[0, a]$ with $f(0) = 0$, $\int_0^a |ff'| \leq \frac{a}{2} \int_0^a (f')^2$, with equality exactly for $f = cx$.
5. Let $f \in C^2[0, 1]$ with $f(0) = f(1) = 0$. Prove that $\int_0^1 (f')^2 \leq \left(\int_0^1 (f'')^2\right)^{1/2} \left(\int_0^1 f^2\right)^{1/2}$ and deduce the L^2 version of the Landau inequality.